

```

:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRUCTINFO :clause {OPEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 17
00 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :cons
tbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof fal
se :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_seria
l 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsiz
e -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRUCTINFO :clause {ALTERNATIVESUBPLAN :subplans ({SUBPLAN :sublinkType 0 :testexpr <> :paramIds <> :plan_id 15
:plan_name SubPlan\ 15 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 15) :args ({VAR :varno 1 :varat
tno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625} {SUBPLAN :sublinkType
e 2 :testexpr {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varle
velsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 16 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :paramIds <> :plan_id 15
:plan_name SubPlan\ 16 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 24.4875 :p
er_call_cost 0.0025})) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1)
:required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_sel
ec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hash
eqoperator 0} {RESTRUCTINFO :clause {OPEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcol
lid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull fa
lse :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security
_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :e
val_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right
_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}))
WARNING: max_parallel_hazard_walker: SubPlan plan_id=15 parallel_safe=0 → reason=correlated_subplan

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRUCTINFO :clause {OPEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1
700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :cons
tbyval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof fal
se :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_seri
al 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsi
ze -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRUCTINFO :clause {ALTERNATIVESUBPLAN :subplans ({SUBPLAN :sublinkType 0 :testexpr <> :paramIds <> :plan_id 1
5 :plan_name SubPlan\ 15 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 15) :args ({VAR :varno 1 :vara
ttno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 24.2625} {SUBPLAN :sublinkType
e 2 :testexpr {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varle
velsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 16 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :paramIds (i 16) :plan_id
16 :plan_name SubPlan\ 16 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 24.4875 :
per_call_cost 0.0025})) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b
1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_se
lec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_has
heqoperator 0} {RESTRUCTINFO :clause {OPEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varco
llid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 1700 :consttypmod -1 :constcollid 0 :constlen -1 :constbyval false :constisnull f
alse :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security
_level 1 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :
eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :righ
t_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPAGATED_RESTRICTED

WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPAGATED_SAFE

WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTe false
:isReturn false
:ceList <>
:relList <>
:reltable (
{RANGEBENTRY
:alias <>
:eref
{ALIAS
:aliasname part
:colnames ("p_partkey" "p_name" "p_mfg" "p_brand" "p_type" "p_size"
"p_container" "p_retailprice" "p_comment")
}
:rtkind 0
:reld 17169
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGEBENTRY
:alias <>
:eref
{ALIAS
:aliasname supplier
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtkind 0
:reld 17181
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:hasGroupRTe false
:lateral false
:inFromCl true
:securityQuals ((
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
}
:location -1
}
}
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:sublinkType 0
:testexpr <>

```

supplier.s_acctbal > 0

```

:paramIds <>
:plan_id 10
:plan_name SubPlan\ 10
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 9)
:args (
  {VAR
    :varno 2
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 4
    :location -1
  }
)
:startup_cost 0
:per_call_cost 24.2625
}
(SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 2
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 4
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 10
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
}
:location -1
}
:paramIds (i 10)
:plan_id 11
:plan_name SubPlan\ 11
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 24.4875
:per_call_cost 0.0025
}
)
}
}
(RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
)
:rtekind 0
:relid 17191
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 3
    :location -1
  }
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 100 0 0 0 0 0 0 ]
  }
}
)
:location -1
}
(ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramIds <>
:plan_id 14
:plan_name SubPlan\ 14
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true

```

n.n_nationkey =
supplier.s_nationkey

ps_availqty > 100

```

:parallel_safe false
:setParam <>
:parParam (i 11)
:args (
  {VAR
    :varno 3
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 2
    :location -1
  }
)
:startup_cost 0
:per_call_cost 121512.00000000003
}
{SUBPLAN
:sublinkType 2
:testexpr
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 3
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 2
        :location -1
      }
      {PARAM
        :paramkind 1
        :paramid 14
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location -1
      }
    )
    :location -1
  }
:paramids (i 14)
:plan_id 17
:plan_name SubPlan\ 17
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam <>
:args <>
:startup_cost 121502.62500000001
:per_call_cost 0.0025
}
)
}
))
}
{RANGETBLENTY
:alias <>
:eref
  {ALIAS
    :aliasname nation
    :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
  }
:rtekind 0
:reldid 17158
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
  {ALIAS
    :aliasname region
    :colnames ("r_regionkey" "r_name" "r_comment")
  }
:rtekind 0
:reldid 17164
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :reldid 17169
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 10 12 13)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :reldid 17181
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11 12 13 14)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :reldid 17191
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 11)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :reldid 17158
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10)
    :insertedCols (b)
    :updatedCols (b)
  }
)

```

s.s_suppkey =
partsupp.ps_suppkey

```

}
{RTEPERMISSIONINFO
:relid 17164
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 9)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
)
}
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 283
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 295
}
)
:location 293
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 318
}
{VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location 330
}
)
:location 328
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 353
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 362
:constvalue 4 [ 15 0 0 0 0 0 0 0 ]
}
)
:location 360
}
{OEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE

```

Now main query part

part.p_partkey =
partsupp.ps_partkey

supplier.s_suppkey =
partsupp.ps_suppkey

part.p_size = 15

```

:arg
{VAR
:varno 1
:varattno 5
:vartype 1043
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 377
}
:resulttype 25
:resulttypmod -1
:resultcollid 100
:relabelformat 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 389
:constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ]
}
)
:location 384
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 410
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 424
}
}
)
:location 422
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 448
}
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location 462
}
}
)
:location 460
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 5
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location 493
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 509
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
)
:location 507
}
{OPEXPR
:opno 1752
:opfuncid 1718
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0

```

part.p_type LIKE
'%BRASS'

supplier.s_nationkey =
nation.n_nationkey

nation.n_regionkey =
region.r_regionkey

region.r_name =
'EUROPE'

```

:args (
  {VAR
    :varno 3
    :varattno 4
    :vartype 1700
    :vartypmod 983046
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 4
    :location 530
  }
  {SUBPLAN
    :sublinkType 4
    :testexpr <>
    :paramIds <>
    :plan_id 9
    :plan_name SubPlan\ 9
    :firstColType 1700
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable false
    :unknownEqFalse false
    :parallel_safe false
    :setParam <>
    :parParam (i 0)
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 812
      }
    )
    :startup_cost 0
    :per_call_cost 48604936345.600006
  }
)
:location 544
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 6
      :vartype 1700
      :vartypmod 983046
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 6
      :location 51
    }
  }
  :resno 1
  :resname s_acctbal
  :ressortgroupref 1
  :resorigtbl 17181
  :resorigcol 6
  :resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 70
}
:resno 2
:resname s_name
:ressortgroupref 3
:resorigtbl 17181
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 86
}
:resno 3
:resname n_name
:ressortgroupref 2
:resorigtbl 17158
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 102
}
:resno 4
:resname p_partkey
:ressortgroupref 4
:resorigtbl 17169
:resorigcol 1
:resjunk false
}
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3

```

partsupp.ps_supplycost
= (subquery)

the actual contents of
the subquery are
stored in the plan tree
for SubPlan 9, not
inline in the outer query
anymore.

```

:varstype 1042
:varystmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 121
}
:resno 5
:resname p_mfgr
:ressortgroupref 0
:resorigtbl 17169
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 3
:varstype 1043
:varystmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 137
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 5
:varstype 1042
:varystmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 156
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 7
:varstype 1043
:varystmod 105
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 7
:location 173
}
:resno 8
:resname s_comment
:ressortgroupref 0
:resorigtbl 17181
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause {
{SORTGROUPCLAUSE
:t1SortGroupRef 1
:eqop 1752
:sortop 1756
:reverse_sort true
:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:t1SortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:t1SortGroupRef 3
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:t1SortGroupRef 4
:eqop 96
:sortop 97
:reverse_sort false
:nulls_first false
:hashable true
}
}
:limitOffset <>
:limitCount
{CONST
:consttype 20
:constystmod -1
:constcollid 0
:constlen 8
:constbval true
:constisnull false
:location -1
:constvalue 8 [ 100 0 0 0 0 0 0 0 ]
}
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0

```



```
-1} {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1}) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 5 :vartype 1043 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 5 :location -1} {VAR :varno 2 :varattno 7 :vartype 1043 :vartypmod 105 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 5 :vartype 1043 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 5 :location -1} {VAR :varno 2 :varattno 7 :vartype 1043 :vartypmod 105 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1})
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 5 :vartype 1043 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 5 :location -1} {VAR :varno 2 :varattno 7 :vartype 1043 :vartypmod 105 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} {VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {VAR :varno 4 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 3 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 5 :vartype 1043 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 5 :location -1} {VAR :varno 2 :varattno 7 :vartype 1043 :vartypmod 105 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1})
QUERY PLAN
Limit (cost=7120623197993.51..7120623197993.52 rows=1 width=271) (actual time=34676.819..34676.844 rows=100.00 loops=1)
Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment
Buffers: shared hit=6351847
-> Sort (cost=7120623197993.51..7120623197993.52 rows=1 width=271) (actual time=34676.819..34676.836 rows=100.00 loops=1)
Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment
Sort Key: supplier.s_acctbal DESC, nation.n_name, supplier.s_name, part.p_partkey
Sort Method: quicksort Memory: 77kB
Buffers: shared hit=6351847
-> Hash Join (cost=3586.00..7120623197993.50 rows=1 width=271) (actual time=13664.210..34676.396 rows=189.00 loops=1)
Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment
Hash Cond: ((partsupp.ps_supplycost = (SubPlan 9)) AND (partsupp.ps_partkey = part.p_partkey))
Buffers: shared hit=6351838
-> Nested Loop (cost=0.00..48604956119.52 rows=291 width=251) (actual time=5.126..7935.083 rows=70201.00 loops=1)
Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, partsupp.ps_partkey, partsupp.ps_supplycost, nation.n_name
Join Filter: (supplier.s_supkey = partsupp.ps_supkey)
Rows Removed by Join Filter: 62190886
Buffers: shared hit=9470
-> Seq Scan on public.partsupp (cost=0.00..48604813792.00 rows=198085 width=14) (actual time=1.441..385.514 rows=70201.00 loops=1)
Output: partsupp.ps_partkey, partsupp.ps_supkey, partsupp.ps_availqty, partsupp.ps_supplycost, partsupp.ps_comment
Filter: ((partsupp.ps_availqty > 100) AND (ANY (partsupp.ps_supkey = (hashed SubPlan 17).col1)))
Rows Removed by Filter: 329799
Buffers: shared hit=8906
SubPlan 17
-> Seq Scan on public.supplier_s_1 (cost=0.00..121499.50 rows=1250 width=4) (actual time=0.049..1.304 rows=887.00 loops=1)
Output: s_1.s_supkey
Filter: ((s_1.s_acctbal > '0'::numeric) AND (ANY (s_1.s_nationkey = (hashed SubPlan 16).col1)) AND (s_1.s_acctbal > '0'::numeric))
Rows Removed by Filter: 4113
Buffers: shared hit=114
SubPlan 16
-> Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.028..0.032 rows=5.00 loops=1)
Output: n_3.n_nationkey
Hash Cond: (n_3.n_regionkey = r_3.r_regionkey)
Buffers: shared hit=2
-> Seq Scan on public.nation_n_3 (cost=0.00..11.70 rows=170 width=8) (actual time=0.007..0.009 rows=25.00 loops=1)
Output: n_3.n_nationkey, n_3.n_name, n_3.n_regionkey, n_3.n_comment
Buffers: shared hit=1
-> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.014..0.014 rows=1.00 loops=1)
Output: r_3.r_regionkey
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=1
-> Seq Scan on public.region_r_3 (cost=0.00..12.12 rows=1 width=4) (actual time=0.008..0.009 rows=1.00 loops=1)
Output: r_3.r_regionkey
Filter: (r_3.r_name = 'EUROPE'::bpchar)
Rows Removed by Filter: 4
Buffers: shared hit=1
-> Materialize (cost=0.00..121528.61 rows=7 width=245) (actual time=0.000..0.031 rows=887.00 loops=70201)
Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, supplier.s_supkey, nation.n_name
Storage: Memory Maximum Storage: 196kB
Buffers: shared hit=564
-> Nested Loop (cost=0.00..121528.58 rows=7 width=245) (actual time=0.052..5.978 rows=887.00 loops=1)
Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, supplier.s_supkey, nation.n_name
Join Filter: (supplier.s_nationkey = nation.n_nationkey)
Rows Removed by Join Filter: 3548
Buffers: shared hit=564
-> Nested Loop (cost=0.00..25.95 rows=1 width=108) (actual time=0.009..0.030 rows=5.00 loops=1)
Output: nation.n_nationkey, nation.n_nationkey
Join Filter: (nation.n_regionkey = region.r_regionkey)
Rows Removed by Join Filter: 20
Buffers: shared hit=2
-> Seq Scan on public.region (cost=0.00..12.12 rows=1 width=4) (actual time=0.003..0.010 rows=1.00 loops=1)
Output: region.r_regionkey, region.r_name, region.r_comment
Filter: (region.r_name = 'EUROPE'::bpchar)
Rows Removed by Filter: 4
Buffers: shared hit=1
-> Seq Scan on public.nation (cost=0.00..11.70 rows=170 width=112) (actual time=0.004..0.014 rows=25.00 loops=1)
Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment
Buffers: shared hit=1
-> Seq Scan on public.supplier (cost=0.00..121487.00 rows=1250 width=145) (actual time=0.008..1.129 rows=887.00 loops=5)
Output: supplier.s_supkey, supplier.s_name, supplier.s_address, supplier.s_nationkey, supplier.s_phone, supplier.s_acctbal, supplier.s_comment
Filter: ((supplier.s_acctbal > '0'::numeric) AND (ANY (supplier.s_nationkey = (hashed SubPlan 11).col1)))
Rows Removed by Filter: 4113
Buffers: shared hit=562
SubPlan 11
-> Hash Join (cost=12.14..24.48 rows=1 width=4) (actual time=0.016..0.020 rows=5.00 loops=1)
Output: n_2.n_nationkey
Hash Cond: (n_2.n_regionkey = r_2.r_regionkey)
Buffers: shared hit=2
-> Seq Scan on public.nation_n_2 (cost=0.00..11.70 rows=170 width=8) (actual time=0.001..0.003 rows=25.00 loops=1)
Output: n_2.n_nationkey, n_2.n_name, n_2.n_regionkey, n_2.n_comment
Buffers: shared hit=1
-> Hash (cost=12.12..12.12 rows=1 width=4) (actual time=0.007..0.008 rows=1.00 loops=1)
Output: r_2.r_regionkey
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=1
-> Seq Scan on public.region_r_2 (cost=0.00..12.12 rows=1 width=4) (actual time=0.002..0.004 rows=1.00 loops=1)
Output: r_2.r_regionkey
Filter: (r_2.r_name = 'EUROPE'::bpchar)
Rows Removed by Filter: 4
Buffers: shared hit=1
-> Hash (cost=3580.00..3580.00 rows=466 width=30) (actual time=13413.548..13413.554 rows=362.00 loops=1)
Output: part.p_partkey, part.p_mfgr
Buckets: 1024 Batches: 1 Memory Usage: 30kB
Buffers: shared hit=3185526
-> Seq Scan on public.part (cost=0.00..3580.00 rows=466 width=30) (actual time=0.052..15.230 rows=362.00 loops=1)
Output: part.p_partkey, part.p_mfgr
Filter: ((part.p_size = 15) AND ((part.p_type)::text ~ '%BRASS'::text))
Rows Removed by Filter: 99638
Buffers: shared hit=2800
SubPlan 9
-> Aggregate (cost=48604936345.59..48604936345.60 rows=1 width=32) (actual time=36.981..36.981 rows=1.00 loops=721)
Output: min(partsupp_1.ps_supplycost)
Buffers: shared hit=6340288
-> Nested Loop (cost=0.00..48604936345.59 rows=1 width=6) (actual time=26.719..36.979 rows=0.71 loops=721)
Output: partsupp_1.ps_supplycost
Join Filter: (nation_1.n_regionkey = region_1.r_regionkey)
Buffers: shared hit=6340288
-> Nested Loop (cost=0.00..48604936333.45 rows=1 width=10) (actual time=26.716..36.976 rows=0.71 loops=721)
Output: partsupp_1.ps_supplycost, nation_1.n_regionkey
Join Filter: (supplier_1.s_nationkey = nation_1.n_nationkey)
Rows Removed by Join Filter: 17
Buffers: shared hit=6339774
-> Nested Loop (cost=0.00..4860493619.63 rows=1 width=10) (actual time=26.711..36.969 rows=0.71 loops=721)
Output: partsupp_1.ps_supplycost, supplier_1.s_nationkey
Join Filter: (partsupp_1.ps_supkey = supplier_1.s_supkey)
```